

Predict Used Car prices in India

Problem Statement:

The goal is to build a machine learning model that accurately estimates the resale price of used cars (Maruti, Tata, Mahindra) based on:

- Brand (categorical)
- Age (years)
- Kilometers Driven (km)

Use Case:

For Buyers:

- Avoid overpaying for used cars.
- Compare prices across brands/age/km combinations.

For Sellers:

- Set competitive resale prices.
- Negotiate better deals using data-backed insights.

For Businesses:

- Used car dealerships can automate pricing.
- Online platforms (e.g., OLX, Spinny) can integrate this model for price suggestions.

Data Set Explanation:

1. Features (Independent Variables)

Column	Description	Data Type	Range/Values
car_name	Brand of the car	Categorical	Maruti, Tata, Mahindra
age_years	Age of the car (in years)	Numerical	1–10 years
kms_driven	Total kilometers driven	Numerical	5,000–100,000 km

2. Target (Dependent Variable)

Column	Description	Data Type	Range (₹)
price	Resale price of the car	Numerical	70,000–5,00,000

Talking: kamal sir sir

ML with Kamal Sir - 8369084928

P4.py

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
from sklearn.neighbors import KNeighborsRegressor
from pickle import dump

data = pd.read_csv("used_car_prices.csv")
print(data)

print(data.isnull().sum())

features = data.drop("price", axis="columns")
target = data["price"]
nfeatures = pd.get_dummies(features)
print(nfeatures)
mms = MinMaxScaler()
sfeatures = mms.fit_transform(nfeatures.values)

k = int(len(data) ** 0.5)
if k % 2 == 0:
    k = k + 1

x_train, x_test, y_train, y_test = train_test_split(sfeatures, target)

model = KNeighborsRegressor(n_neighbors=k, metric="euclidean")
model.fit(x_train, y_train)

score = model.score(x_test, y_test)
print(score)

with open("car.pkl", "wb") as f:
    dump(model, f)
    print("model saved")

with open("mms.pkl", "wb") as f:
    dump(mms, f)
    print("mms, saved")
```


P5.py

```
from pickle import load

with open("car.pkl", "rb") as f:
    model = load(f)
    print("model ready")

with open("mms.pkl", "rb") as f:
    mms = load(f)
    print("mms ready")

age = float(input("enter age "))
kms = float(input("enter kms "))
name = int(input("1 for Mahindra, 2 for Maruti and 3 for Tata "))
if name == 1:
    d = [[age, kms, 1, 0, 0]]
elif name == 2:
    d = [[age, kms, 0, 1, 0]]
else:
    d = [[age, kms, 0, 0, 1]]

sd = mms.transform(d)
ans = model.predict(sd)
print(ans)
```